A Geometric View of Model Merging: Quotient Fréchet Averages from Toy Models to LoRA

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Abstract

Model merging combines task-specialized models without additional training, but weight-space averaging is not intrinsically well-defined when parameters have symmetries. Two checkpoints can represent the same or closely related function while occupying different points along a symmetry orbit; averaging arbitrary representatives can therefore produce a degraded merge. We formulate model merging as Fréchet averaging on a Riemannian parameter geometry, and argue that in the presence of symmetries the appropriate object is a Fréchet mean on the quotient space of parameterizations modulo symmetry. We work through a two-parameter linear model with a scaling symmetry, where the quotient geometry is closed form: quotient GeoMerge reduces to averaging the invariant predictors, while Euclidean and Fisher-weighted parameter averaging can collapse for gauge-equivalent checkpoints. We then instantiate the same principle for LoRA adapters and report proof-of-concept ViT-B/32 merging results, where quotient-aware GeoMerge improves over strong alignment-based LoRA merging baselines.

1. Introduction

Model merging aims to combine multiple trained networks into a single model that preserves the union of their capabilities while avoiding additional end-to-end training. This objective is increasingly salient because modern workflows routinely produce families of related models: different seeds and hyperparameters, domain- or task-specialists, safety patches, preference updates, and personalized variants. Existing methods include data-dependent approaches such as Fisher merging (Matena and Raffel, 2022), RegMean (Jin et al., 2025), and MaTS (Tam et al., 2024), as well as weight-only approaches such as model soups (Wortsman et al., 2022), Task Arithmetic (Ilharco et al., 2023), and TIES (Yadav et al., 2023). We focus on the weight-only setting, where only checkpoints are available at merge time.

The difficulty is that neural-network parameters are not canonical coordinates for functions. Modern architectures have large symmetry groups under which distinct parameter settings implement exactly the same input-output map. Classic examples include neuron permutations (Hecht-Nielsen, 1990) and rescalings in positively homogeneous networks (Dinh et al., 2017). Transformers exhibit richer continuous symmetries, including high-dimensional $\mathrm{GL}(h)$ symmetries in attention blocks (da Silva et al., 2025; Zhang et al., 2025). Low-rank updates introduce their own gauge freedoms: a LoRA factorization may change while the represented update remains fixed. If two checkpoints differ partly by such a symmetry, their Euclidean midpoint is an average of arbitrary representatives rather than an average of intrinsic models. The merge can step off the low-loss set, or even depend on which equivalent parameterization was chosen before merging.

This perspective connects several lines of work. Model soups and task-vector methods show that simple weight-space operations can work when checkpoints lie in a shared basin or approximately linearly connected region (Wortsman et al., 2022; Ilharco et al., 2023; Garipov et al., 2018; Draxler et al., 2018). Other merging methods introduce data-dependent metrics, interference mitigation, or alignment stages (Matena and Raffel, 2022; Jin et al., 2025; Tam et al., 2024; Yadav et al., 2023). In particular, KnOTS and Core Space show empirically that LoRA merging benefits from aligning low-rank updates before applying standard merge rules (Stoica et al., 2025; Panariello et al., 2025). We address this by making the geometric object being aligned explicit: when parameterizations have symmetries, the merge should be defined on the quotient space of intrinsic models or updates, and computed as a quotient Fréchet mean.

We propose **GeoMerge**: a geometric formulation of model

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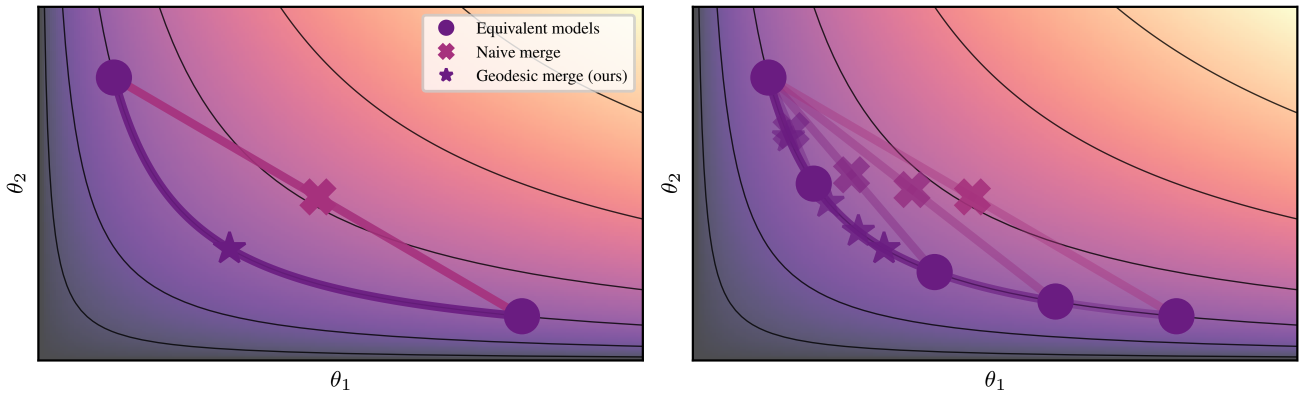

Figure 1. Visual comparison of symmetry-unaware and symmetry-aware merging. *Left:* Euclidean averaging can leave the low-loss orbit because it measures straight-line proximity in the wrong geometry. *Right:* representative-level merging is ambiguous: equivalent reparameterizations of the same models can produce different Euclidean averages. Quotient GeoMerge instead averages intrinsic equivalence classes, yielding a merge that is invariant to the chosen representatives.

merging as Fréchet averaging on a Riemannian parameter space, with quotient geometry used when symmetries make representatives non-identifiable. This view turns alignment from a heuristic preprocessing step into the orbit-minimization step of a well-defined quotient average. Our contributions are:

- We formulate model merging as Fréchet averaging and show that symmetries require quotient rather than representative-level averages.
- We analyze a two-parameter linear model with a scaling symmetry, where quotient GeoMerge is closed form and reduces to averaging invariant predictors.
- We instantiate the same principle for LoRA adapters and provide ViT-B/32 proof-of-concept results showing gains over alignment-based LoRA merging baselines.

2. Merging as quotient Fréchet averaging

Let $x_1, \dots, x_T$ be task-specific models or updates with weights $w_i \geq 0$ and $\sum_i w_i = 1$. Given a metric space or Riemannian manifold $(\mathcal{M}, d_{\mathcal{M}})$, a Fréchet mean is any minimizer

$$\mu^\star \in \arg\min_{\mu \in \mathcal{M}} \frac{1}{2} \sum_{i=1}^{T} w_i d_{\mathcal{M}}(\mu, x_i)^2. \tag{1}$$

Euclidean weight averaging is recovered when $\mathcal{M}$ is a flat vector space equipped with the Euclidean distance. More generally, a merge is defined only after specifying the space of objects being averaged and the distance used to compare them. Fisher merging replaces Euclidean squared distances with local quadratic approximations (defined by the Fisher information at each checkpoint), while alignment-based methods change representatives before applying an averaging rule. GeoMerge unifies these perspectives by computing the average on the intrinsic geometry of the objects being merged, rather than in arbitrary coordinates.

If parameters have a symmetry, the intrinsic space of models is naturally a quotient rather than the raw parameter space. Let a group $\mathcal{G}$ act on a total space $\overline{\mathcal{M}}$, and suppose $\bar{\theta}$ and $\bar{\theta} \cdot g$ represent the same intrinsic model for every $g \in \mathcal{G}$. Then the quotient space is $\mathcal{M} = \overline{\mathcal{M}}/\mathcal{G}$, with points $[\bar{\theta}] = \{\bar{\theta} \cdot G : G \in \mathcal{G}\}$. To define a geometry on this quotient, the metric on $\overline{\mathcal{M}}$ must be compatible with the symmetry. In the cleanest case, the group action is isometric: applying the same symmetry transformation to two representatives preserves their distance. Then the quotient distance can be written as an orbit-minimum,

$$d_{\mathcal{M}}([\bar{\theta}], [\bar{\phi}]) = \inf_{G \in \mathcal{G}} d_{\overline{\mathcal{M}}}(\bar{\theta}, \bar{\phi} \cdot G). \tag{2}$$

Equivalently, one first aligns representatives along their symmetry orbits and then compares them in directions that change the intrinsic model rather than the gauge.

Plugging this distance into (1) yields a merge that depends only on the orbits (i.e. quotient points $[\bar{\theta}_i]$), not on the arbitrary representatives chosen in parameter space.

3. A toy geometry where GeoMerge is closed form

We analyze a minimal model with a scaling symmetry to make the symmetry-induced failure modes of naive parameter averaging concrete and to illustrate GeoMerge as quotient/geodesic merging in closed form.

A two-layer linear network with a scaling gauge. We start with the smallest setting where a continuous architectural symmetry already makes "parameter-space averaging" ill-posed. Consider a scalar two-layer linear network

$$f_\theta(x) = \theta_2 \theta_1 x, \qquad \theta = (\theta_1, \theta_2) \in \overline{\mathcal{M}} := (\mathbb{R}^\star)^2, \tag{3}$$

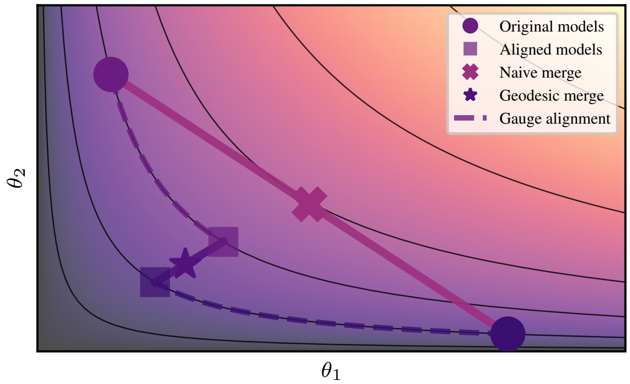

Figure 2. Geodesic merging in the toy model. Each checkpoint has infinitely many gauge-equivalent representatives along a symmetry orbit (dashed). Naive ambient averaging depends on the chosen representatives and can move to the wrong orbit. GeoMerge first aligns representatives along their orbits, then averages along a horizontal geodesic, yielding a quotient-consistent merge.

where $\mathbb{R}^\star = \mathbb{R} \setminus \{0\}$. This parametrization has a multiplicative gauge symmetry: for any $a \in \mathbb{R}^\star$, $(\theta_1, \theta_2) \cdot a = (a\theta_1, \theta_2/a)$. Since $f_{\theta \cdot a} \equiv f_\theta$, the predictor $\beta(\theta) := \theta_1\theta_2$ is invariant under the action. Thus the intrinsic object is the orbit $[\theta] \in \mathcal{M} := \overline{\mathcal{M}}/\mathbb{R}^\star$.

A symmetry-invariant metric. To respect the scaling symmetry, we equip $\overline{\mathcal{M}}$ with the scalar specialization of the symmetry-invariant metric for (G, H)-type parameters (da Silva et al., 2025):

$$g_{(\theta_1,\theta_2)}\left[(\eta_1, \eta_2), (\zeta_1, \zeta_2)\right] = \theta_2^2\eta_1\zeta_1 + \theta_1^2\eta_2\zeta_2, \tag{4}$$

with $(\eta_1, \eta_2), (\zeta_1, \zeta_2) \in T_\theta\overline{\mathcal{M}} \cong \mathbb{R}^2$. A direct calculation shows that the scaling action acts by isometries under g, so the quotient $\mathcal{M}$ inherits a well-defined metric.

Vertical vs. horizontal directions. Let $a = \exp(s)$ and differentiate the group action at $s = 0$ to obtain the vertical, orbit-changing direction $v(\theta) := \frac{d}{ds}\big|_{s=0} (\theta \cdot e^s) = (\theta_1, -\theta_2)$. The horizontal space is the g-orthogonal complement of $\mathrm{span}\{v(\theta)\}$. Thus $\eta = (\eta_1, \eta_2)$ is horizontal iff

$$g_\theta[\eta, v(\theta)] = 0 \quad \Longleftrightarrow \quad \frac{\eta_1}{\theta_1} = \frac{\eta_2}{\theta_2}. \tag{5}$$

Intuitively, horizontal motion changes the intrinsic predictor $\beta = \theta_1\theta_2$, while vertical motion changes only the representative, i.e. the gauge.

Horizontal geodesics make the predictor evolve linearly. In this toy geometry, geodesics admit a closed form. Writing $\gamma(t) = (\theta_1(t), \theta_2(t))$ with initial velocity $\dot\gamma(0) = (\eta_1(0), \eta_2(0))$, one obtains

$$\begin{aligned} \theta_1(t) &= \theta_1(0)\sqrt{1 + 2\eta_1(0)\theta_1(0)^{-1}t}, \\ \theta_2(t) &= \theta_2(0)\sqrt{1 + 2\eta_2(0)\theta_2(0)^{-1}t}. \end{aligned} \tag{6}$$

If the initial velocity is horizontal, (5) implies $\frac{\eta_1(0)}{\theta_1(0)} = \frac{\eta_2(0)}{\theta_2(0)} =: B$. Then the ratio $\theta_1(t)/\theta_2(t) = \theta_1(0)/\theta_2(0)$ is fixed, and

$$\beta(t) := \theta_1(t)\ \theta_2(t) = \theta_1(0)\ \theta_2(0)\ (1 + 2Bt). \tag{7}$$

The quotient geometry has removed the non-identifiable scale degree of freedom; the only remaining coordinate is the predictor β. And thus, once two points are gauge-aligned (so a horizontal geodesic connects them), the intrinsic interpolation is simply linear in predictor space.

Quotient distance reduces to predictor distance after alignment. Because the action is isometric, the quotient distance can be written as an orbit-minimum:

$$d_{\mathcal{M}}([\theta], [\kappa]) = \min_{a \in \mathbb{R}^\star} d_{\overline{\mathcal{M}}}(\theta, \kappa \cdot a). \tag{8}$$

In this model, choosing a so that θ and $\kappa \cdot a$ lie on the same horizontal geodesic, i.e. so they share the same ratio $\frac{\theta_1}{\theta_2}$ and lie in the same connected component, yields a closed-form distance that depends only on the invariant predictors:

$$d_{\mathcal{M}}([\theta], [\kappa]) \propto |\beta(\theta) - \beta(\kappa)| = |\theta_1\theta_2 - \kappa_1\kappa_2|. \tag{9}$$

GeoMerge becomes "average the predictors". Given checkpoints $\theta^{(1)}, \ldots, \theta^{(T)}$, define $\beta_i := \beta(\theta^{(i)}) = \theta_1^{(i)}\theta_2^{(i)}$. With (9), the quotient Fréchet objective becomes

$$\begin{aligned} \mu^\star &= \arg\min_{[\mu] \in \mathcal{M}} \frac{1}{2}\sum_{i=1}^{T} d_{\mathcal{M}}([\mu], [\theta^{(i)}])^2 \\ &= \arg\min_{\beta \in \mathbb{R}} \frac{1}{2}\sum_{i=1}^{T} (\beta - \beta_i)^2. \end{aligned} \tag{10}$$

Therefore the merged intrinsic predictor is $\beta^\star = \frac{1}{T}\sum_{i=1}^{T}\beta_i$. Mapping back to parameters corresponds to choosing any representative $\mu = (\mu_1, \mu_2)$ with $\mu_1\mu_2 = \beta^\star$. This last step is only a gauge choice.

Why this toy problem matters. Take two checkpoints that are different representatives of *the same function*, e.g., $\theta^{(2)} = \theta^{(1)} \cdot (-1) = (-\theta_1^{(1)}, -\theta_2^{(1)})$. Naïve Euclidean averaging gives $\frac{\theta^{(1)} + \theta^{(2)}}{2} = (0, 0)$, which lies outside $\overline{\mathcal{M}}$ and corresponds to the zero predictor. Fisher-weighted averaging exhibits the same pathology in this example: the Fisher can coincide for the two representatives while the parameters cancel, causing the merged parameters to collapse to $(0, 0)$. The bad merge results from averaging in the wrong coordinate system. GeoMerge works on the quotient, where, in this case, $[\theta^{(1)}] = [\theta^{(2)}]$, and hence $d_{\mathcal{M}}([\theta^{(1)}], [\theta^{(2)}]) = 0$. The Fisher derivation is included in Section B.

What does the toy model show? It reveals three points that persist in larger networks. First, the failure of Euclidean merging is not caused by task conflict: even two gauge-equivalent checkpoints representing the same predictor can average to a bad model. Second, changing the metric alone is not enough unless the averaging procedure also respects the symmetry; the merge must be invariant to choice of representative on each orbit. Third, alignment methods can be understood geometrically as approximations to quotient averaging: they choose comparable representatives before averaging directions that change the intrinsic model.

The closed form above is special to this toy geometry. Here the quotient removes the scale degree of freedom and leaves a flat one-dimensional predictor coordinate, so quotient GeoMerge reduces to ordinary averaging of β. In higher-rank LoRA and transformer settings, the quotient geometry is curved, and curvature affects both distances and averages. Thus the toy model should not be read as saying that GeoMerge is always predictor averaging; rather, it isolates the symmetry obstruction in a setting where the correct quotient average can be written explicitly.

4. From the toy model to LoRA

The toy example is intentionally small, but it isolates the same obstruction that appears in low-rank adapters: the represented update is identifiable, while its factorization is not. A rank-r LoRA update can be represented as a dense matrix $\Delta = GH^\top$, but the factors are not unique. We use the compact quotient representation

$$\begin{aligned} \theta &= [U, B, V] \in \mathcal{M}_r, \qquad \Delta = UBV^\top, \\ \mathcal{M}_r &= \big(\mathrm{St}(d_{\mathrm{out}}, r) \times \mathrm{SPD}(r) \times \mathrm{St}(d_{\mathrm{in}}, r)\big)/O(r), \end{aligned} \tag{11}$$

so that averaging is performed over represented updates rather than arbitrary LoRA coordinates. The quotient by $O(r)$ identifies different orthogonal gauges of the same low-rank update, just as the toy quotient identifies different scalings of the same scalar predictor. For a practical comparison with KnOTS and Core Space (Stoica et al., 2025; Panariello et al., 2025), we use a quotient-compatible high-rank lift and then compute the quotient Fréchet mean.

Practical LoRA implementation. A trained LoRA update is stored as $\Delta = GH^\top$, but this factorization has the gauge $(G, H) \sim (GA^{-1}, HA^\top)$ for $A \in \mathrm{GL}(r)$. We convert each update to the compact polar representative $\Delta = UBV^\top$ and work with the residual $O(r)$ gauge $(U, B, V) \sim (UO, O^\top BO, VO)$. The metric is the canonical Stiefel metric for U, V and the affine-invariant SPD metric for B. The lift is a direct construction of rank-R quotient points. The quotient Fréchet solver is used only after this lift: in its first outer step, each input gauge is initialized by the Procrustes solve $O_i = \mathrm{polar}(U_i^\top U_p + V_i^\top V_p)$. Later steps keep the gauge variables and update them through gauge-drift horizontalization. The solver averages the resulting horizontal logarithms and updates the Stiefel and SPD factors.

Existing LoRA merging methods often combine two choices: a rank-increasing lift and a merge rule in the lifted coordinates. KnOTS aligns task updates in an SVD-derived coordinate system, while Core Space builds shared left and right bases before merging the induced cores (Stoica et al., 2025; Panariello et al., 2025). GeoMerge separates these choices. The lift supplies a common rank-R representation, while the merge rule is the quotient Fréchet mean on $\mathcal{M}_R$. For each task t, with $\theta_t = [U_t, B_t, V_t] \in \mathcal{M}_r$ and $\Delta_t = U_t B_t V_t^\top$, we construct a lifted point $L_t = [\widehat{U}_t, \widehat{B}_t, \widehat{V}_t] \in \mathcal{M}_R$ and minimize $\frac{1}{2}\sum_t w_t d_{\mathcal{M}_R}(\mu, L_t)^2$.

The lift keeps the task's original subspace and adds directions that explain residual structure from the other tasks. With projectors $P_t^U = I - U_t U_t^\top$ and $P_t^V = I - V_t V_t^\top$, the residual $R_t = \sum_{s \neq t} P_t^U \Delta_s P_t^V$ selects components not already represented by task t. We take leading paired singular directions of R_t as the added columns of $\widehat{U}_t$ and $\widehat{V}_t$, and set $\widehat{B}_t = \mathrm{diag}(B_t, cI_{R-r})$ with $c = (\prod_s \lambda_{\min}(B_s))^{1/T}$. This construction is quotient-defined: Δ_s, P_t^U, and P_t^V do not depend on the $O(r)$ gauge of the input factors, and sign or rotation ambiguity in the added singular directions is absorbed by the $O(R)$ quotient.

As proof of concept, we evaluate on a LoRA-adapted ViT-B/32 setup from prior work (Stoica et al., 2025). The benchmark merges eight specialists fine-tuned from the same base model on Cars, DTD, EuroSAT, GTSRB, MNIST, RESISC, SUN397, and SVHN (Dosovitskiy et al., 2020; Krause et al., 2013; Cimpoi et al., 2014; Helber et al., 2019; Stallkamp et al., 2011; LeCun, 1998; Cheng et al., 2017; Xiao et al., 2016; Netzer et al., 2011). The evaluation is weight-only and gradient-free at merge time. Each specialist consists of a shared ViT-B/32 base together with a task-specific LoRA update. We merge the shared backbone and LoRA parameters while retaining task-specific classification heads, so each dataset evaluates the merged representation with its corresponding head. GeoMerge+lift obtains the best average normalized accuracy among the compared methods: 77.10%, compared with 76.43% for the best Core baseline and 74.02% for the best KnOTS baseline. Detailed results are given in Table 1 in Section 4.

Table 1 compares against strong alignment-based LoRA merging baselines. GeoMerge+lift obtains the best average normalized accuracy among the compared methods: 77.10%, compared with 76.43% for the best Core baseline and 74.02% for the best KnOTS baseline. We also include an ablation that applies TSV+Iso-C after our lift but without quotient Fréchet averaging. This ablation underperforms the best Core lift, suggesting that the gains are not solely

Table 1. Merged-model performance on vision tasks for ViT-B/32 in normalized accuracies (%).

Method	**Space**	**Cars**	**DTD**	**EuroSAT**	**GTSRB**	**MNIST**	**RESISC**	**SUN397**	**SVHN**	**Avg**
TA	Full	81.97	73.72	48.97	42.24	53.12	71.50	97.46	41.25	63.78
TIES	Full	82.37	72.72	49.91	36.62	57.16	69.38	96.92	44.56	63.70
	KnOTS	83.75	74.45	50.36	47.31	67.01	71.79	96.51	50.64	67.73
	Core	84.74	76.46	52.19	50.41	67.36	71.21	96.45	50.18	68.63
Iso-C	Full	80.16	83.03	51.44	74.76	70.72	79.89	98.66	50.20	73.60
	KnOTS	80.33	79.29	57.50	67.60	65.63	79.54	99.26	46.62	71.97
	Core	83.35	84.30	50.13	81.97	71.07	83.46	99.17	53.90	75.92
TSV+Iso-C	Full	79.38	80.38	57.99	65.64	64.22	79.74	98.59	46.49	71.55
	KnOTS	80.81	83.03	58.25	74.34	67.66	79.69	98.54	49.86	74.02
	Core	82.98	85.12	50.95	84.25	71.14	84.39	99.06	53.53	76.43
Ours	TSV+Iso-C on our lift	82.47	84.21	53.75	81.96	72.48	79.88	99.26	53.72	75.97
	GeoMerge+lift	83.48	84.30	53.27	82.14	75.45	82.22	98.76	57.18	**77.10**

explained by the rank lift. These results support the main claim that symmetry-aware averaging improves practical LoRA merging.

Result interpretation. The table provides two controls. Within each Euclidean merge rule, the "Full," "KnOTS," and "Core" rows show how much performance changes when the same arithmetic rule is preceded by a representation change. Across the two "Ours" rows, the representation is fixed to our lift while the averaging rule changes. TSV+Iso-C on our lift reaches 75.97%, below TSV+Iso-C in Core space at 76.43%; replacing that Euclidean rule with the quotient Fréchet mean reaches 77.10%. The gain therefore does not come from the lift alone. It appears only when the lifted updates are averaged with the quotient geometry.

Protocol and cost. All methods in the table are weight-only at merge time. GeoMerge uses the same checkpoints and heads as the baselines, but performs no task-data access, label access, or gradient updates during the merge. Lifted GeoMerge with Cayley retractions and logarithms took around 20 minutes per vision run on a Titan V GPU with a 12-core Intel Xeon W-2133 CPU. The exact exponential/logarithm variant is slower; we use the Cayley implementation for the reported lifted vision result.

Appendix roadmap. The appendix contains the details omitted from the main text: Riemannian and quotient definitions, a closed-form toy problem showing how symmetry can make naive averaging ill-defined, the relation to Fisher merging, implementation details for quotient descent, the high-rank lift, and additional NLI experiments.

5. Conclusion

GeoMerge proposes an alternative, geometric view on model merging by treating it as computing a Fréchet mean under an explicit geometry, rather than an arbitrary average in parameter space. This perspective directly addresses architectural and parameterization symmetries: when models live on orbits of equivalent representations. By operating on the appropriate manifold via orbit alignment and geodesic updates, our merges are symmetry-consistent by construction and grant access to the deeper geometric structure of parameter space. Empirical evidence supports the effectiveness of our proposed method.

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Appendix

A. Riemannian and Quotient Geometry Background

A.1. Riemannian manifolds, distance, and Exp/Log.

A *Riemannian manifold* is a pair $(\mathcal{M}, g)$, where $\mathcal{M}$ is a smooth manifold and the metric g assigns to each $p \in \mathcal{M}$ an inner product $g_p : T_p\mathcal{M} \times T_p\mathcal{M} \to \mathbb{R}$ on the tangent space $T_p\mathcal{M}$ at p. For the tangent vectors ξ, ζ we write $\langle \xi, \zeta \rangle_p := g_p(\xi, \zeta)$ and $\|\xi\|_p := \sqrt{\langle \xi, \xi \rangle_p}$. A *geodesic* is a curve $\gamma : [0, 1] \to \mathcal{M}$ that locally minimizes length. The *geodesic distance* between two points $p, q \in \mathcal{M}$ is

$$d_{\mathcal{M}}(p, q) \ := \inf_{\gamma(0)=p,\, \gamma(1)=q} \int_0^1 \|\dot{\gamma}(t)\|_{\gamma(t)}\, dt.$$

The *exponential map* $\mathrm{Exp}_p : T_p\mathcal{M} \to \mathcal{M}$ maps a tangent vector ξ to the endpoint of the geodesic starting at p with initial velocity ξ: $\mathrm{Exp}_p(\xi) := \gamma_\xi(1)$ where $\gamma_\xi(0) = p$, $\dot{\gamma}_\xi(0) = \xi$. Its (local) inverse is the *logarithm map* $\mathrm{Log}_p : \mathcal{M} \to T_p\mathcal{M}$ satisfying $\mathrm{Exp}_p(\mathrm{Log}_p(q)) = q$. These mappings are later used to obtain the Fréchet mean.

Quotients induced by symmetries. Neural network parameterizations often admit symmetries: many distinct parameter settings encode the same function. A principled way to enforce symmetry-invariance is to work on a *quotient manifold*. Let $\overline{\mathcal{M}}$ be a (total) manifold with a smooth right action of a Lie group $\mathcal{G}$ so that $\Psi : \overline{\mathcal{M}} \times \mathcal{G} \to \overline{\mathcal{M}}$ and $(\bar{p}, G) \mapsto \bar{p} \cdot G$. This action induces an equivalence relation $\bar{p} \sim \bar{q}$ if $\bar{q} = \bar{p} \cdot G$ for some $G \in \mathcal{G}$. The quotient space is $\mathcal{M} := \overline{\mathcal{M}}/\mathcal{G}$; we write $[\bar{p}] \in \mathcal{M}$ for the orbit and $\pi : \overline{\mathcal{M}} \to \mathcal{M}$ for the projection $\pi(\bar{p}) = [\bar{p}]$.

Vertical and horizontal spaces. At a point $\bar{p} \in \overline{\mathcal{M}}$, the *vertical space* $V_{\bar{p}}\overline{\mathcal{M}} \subset T_{\bar{p}}\overline{\mathcal{M}}$ is the tangent space to the orbit through $\bar{p}$, i.e. the set of infinitesimal symmetry directions. If $\overline{\mathcal{M}}$ has a Riemannian metric $\bar{g}$ under which the action is by isometries (metric is symmetry invariant), then we define the *horizontal space* as the orthogonal complement w.r.t the Riemannian metric $\bar{g}$

$$H_{\bar{p}}\overline{\mathcal{M}} \ := \ \left(V_{\bar{p}}\overline{\mathcal{M}}\right)^{\perp} \subset T_{\bar{p}}\overline{\mathcal{M}}.$$

Tangent vectors in the quotient space are set-valued and therefore inconvenient for numerical manipulation. Conveniently, horizontal vectors provide representatives of tangent vectors on the quotient space: each $\xi \in T_{[\bar{p}]}\mathcal{M}$ has a unique *horizontal lift* $\bar{\xi} \in H_{\bar{p}}\overline{\mathcal{M}}$ with $d\pi_{\bar{p}}(\bar{\xi}) = \xi$ (Absil et al., 2008; Boumal, 2023).

Quotient distance as orbit alignment. If $\bar{g}$ is $\mathcal{G}$-invariant, it induces a well-defined metric g on the quotient. The resulting geodesic distance on $\mathcal{M}$ can be written as an *orbit-minimum*:

$$d_{\mathcal{M}}\big([\bar{p}], [\bar{q}]\big) \ = \ \min_{G \in \mathcal{G}} d_{\bar{\mathcal{M}}}\big(\bar{p}, \bar{q} \cdot G\big). \tag{12}$$

We will refer to a minimizer (a symmetry transformation that leads to the smallest geodesic distance)

$$G^\star(\bar{p}; \bar{q}) \in \arg\min_{G \in \mathcal{G}} d_{\bar{\mathcal{M}}}\big(\bar{p}, \bar{q} \cdot G\big)^2 \tag{13}$$

as an *alignment* (or *gauge choice*) of $\bar{q}$ to $\bar{p}$, and write the aligned representative as $\bar{q}^\star := \bar{q} \cdot G^\star(\bar{p}; \bar{q})$. Even when $\bar{q}$ and $\bar{q}^\star$ are different points in $\overline{\mathcal{M}}$, they represent the same quotient point: $[\bar{q}] = [\bar{q}^\star]$. Alignment implies horizontal geodesics: whenever $\bar{p}$ and $\bar{q}$ are aligned, $\mathrm{Log}_{\bar{p}}(\bar{q})$ is horizontal (e.g., Huckemann et al. (2010)), that is, the points are connected by a horizontal geodesic that is always perpendicular to orbits, i.e. $\dot{\gamma}(t) \in H_{\gamma(t)}\mathcal{M}$. This will be a representation of the geodesic on the quotient space (which is an abstract space) that we can work with numerically.

B. Fisher Information Matrix for the Toy Model

We consider the regression model defined by the function $f(x; \boldsymbol{\theta}) = \theta_1\theta_2 x$. The observed data consists of n pairs $\{(x_i, y_i)\}_{i=1}^n$, and we assume additive Gaussian noise with variance σ^2. The model is given by:

$$y_i = \theta_1\theta_2 x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2) \tag{14}$$

The parameter vector is $\boldsymbol{\theta} = [\theta_1, \theta_2]^\top$. The log-likelihood function $\ell(\boldsymbol{\theta})$ for the observations is:

$$\ell(\boldsymbol{\theta}) = -\frac{n}{2}\log\big(2\pi\sigma^2\big) - \frac{1}{2\sigma^2}\sum_{i=1}^n (y_i - \theta_1\theta_2 x_i)^2 \tag{15}$$

The score function is the gradient of the log-likelihood with respect to the parameters, $\nabla_{\boldsymbol{\theta}}\ell(\boldsymbol{\theta})$. The partial derivatives are:

$$\frac{\partial \ell}{\partial \theta_1} = \frac{1}{\sigma^2}\sum_{i=1}^n (y_i - \theta_1\theta_2 x_i)(\theta_2 x_i) \tag{16}$$

$$\frac{\partial \ell}{\partial \theta_2} = \frac{1}{\sigma^2}\sum_{i=1}^n (y_i - \theta_1\theta_2 x_i)(\theta_1 x_i) \tag{17}$$

The Hessian matrix $\mathbf{H}$ consists of the second-order partial derivatives:

$$\frac{\partial^2 \ell}{\partial \theta_1^2} = -\frac{1}{\sigma^2}\sum_{i=1}^n \theta_2^2 x_i^2 \tag{18}$$

$$\frac{\partial^2 \ell}{\partial \theta_2^2} = -\frac{1}{\sigma^2}\sum_{i=1}^n \theta_1^2 x_i^2 \tag{19}$$

$$\frac{\partial^2 \ell}{\partial \theta_1 \partial \theta_2} = \frac{1}{\sigma^2}\sum_{i=1}^n (y_i x_i - 2\theta_1\theta_2 x_i^2) \tag{20}$$

The Fisher Information Matrix (FIM), $\mathcal{I}(\boldsymbol{\theta})$, is defined as the negative expectation of the Hessian matrix:

$$\mathcal{I}(\boldsymbol{\theta}) = -E\left[\mathbf{H}\right] \tag{21}$$

We compute the expectation of the mixed partial derivative term using the relation $E[y_i] = \theta_1\theta_2 x_i$:

$$E\left[\frac{\partial^2 \ell}{\partial\theta_1\partial\theta_2}\right] = \frac{1}{\sigma^2}\sum_{i=1}^{n}(\theta_1\theta_2 x_i^2 - 2\theta_1\theta_2 x_i^2) = -\frac{1}{\sigma^2}\sum_{i=1}^{n}\theta_1\theta_2 x_i^2 \tag{22}$$

Let $S_{xx} = \sum_{i=1}^{n} x_i^2$. The Fisher Information Matrix is therefore:

$$\mathcal{I}(\boldsymbol{\theta}) = \frac{S_{xx}}{\sigma^2}\begin{bmatrix}\theta_2^2 & \theta_1\theta_2 \\ \theta_1\theta_2 & \theta_1^2\end{bmatrix} \tag{23}$$

Note that $\det(\mathcal{I}(\boldsymbol{\theta})) = 0$, indicating that the matrix is singular and the parameters θ_1 and θ_2 are unidentifiable as already pointed out.

C. Low-rank LoRA Quotient Details

C.1. Low-rank updates and their symmetries.

A common PEFT primitive is a rank-r update to a weight matrix. We consider updates of the form

$$\Delta\boldsymbol{W} \in \mathbb{R}^{d_1\times d_2}, \quad \operatorname{rank}(\Delta\boldsymbol{W}) = r\,,$$

that admit several factorizations. We consider the following (Hu et al., 2022; Mishra et al., 2013):

(A) Standard factorization ($\mathrm{GL}(r)$ symmetry). $\Delta\boldsymbol{W}$ is usually trained and stored as a factorization $\Delta\boldsymbol{W} = \boldsymbol{G}\boldsymbol{H}^\top$ with $\boldsymbol{G} \in \mathbb{R}_\star^{d_1\times r}$ and $\boldsymbol{H} \in \mathbb{R}_\star^{d_2\times r}$ ($_\star$ indicating full column rank). Then, for any $\boldsymbol{A} \in \mathrm{GL}(r)$, $(\boldsymbol{G}\boldsymbol{A}^{-1})(\boldsymbol{H}\boldsymbol{A}^\top)^\top = \boldsymbol{G}\boldsymbol{H}^\top$ and we define the equivalence relationship

$$(\boldsymbol{G},\boldsymbol{H}) \sim (\boldsymbol{G}\boldsymbol{A}^{-1},\, \boldsymbol{H}\boldsymbol{A}^\top).$$

(B) Polar factorization ($\mathrm{O}(r)$ symmetry). A rank-r matrix admits an equivalent polar factorization

$$\Delta\boldsymbol{W} = \boldsymbol{U}\boldsymbol{B}\boldsymbol{V}^\top \tag{24}$$

where, $\boldsymbol{U} \in \mathrm{St}(d_1, r)$, $\boldsymbol{V} \in \mathrm{St}(d_2, r)$, $\boldsymbol{B} \in \mathbb{S}_{++}(r)$, and $\mathrm{St}(d, r) = \{\boldsymbol{U} \in \mathbb{R}^{d\times r} : \boldsymbol{U}^\top\boldsymbol{U} = \boldsymbol{I}_r\}$ is the Stiefel manifold and $\mathbb{S}_{++}(r)$ the SPD manifold of rank r. This representation has the (smaller) orthogonal symmetry group: for any $\boldsymbol{O} \in \mathrm{O}(r)$,

$$(\boldsymbol{U},\boldsymbol{B},\boldsymbol{V}) \sim (\boldsymbol{U}\boldsymbol{O},\, \boldsymbol{O}^\top\boldsymbol{B}\boldsymbol{O},\, \boldsymbol{V}\boldsymbol{O}),$$

since $(\boldsymbol{U}\boldsymbol{O})(\boldsymbol{O}^\top\boldsymbol{B}\boldsymbol{O})(\boldsymbol{V}\boldsymbol{O})^\top = \boldsymbol{U}\boldsymbol{B}\boldsymbol{V}^\top$. This $\mathrm{O}(r)$-quotient viewpoint is particularly convenient computationally because geometric primitives like Exp, and Log either admit exact expressions or accurate numerical routines (Edelman et al., 1998; Mataigne et al., 2024).

The quotient geometry of the polar factorization. We follow the construction from Mishra et al. (2013). On $\mathbb{S}_{++}$ we will use the affine-invariant metric:

$$g_{\boldsymbol{B}}(\boldsymbol{\zeta_B},\boldsymbol{\eta_B}) = \langle\boldsymbol{\zeta_B},\boldsymbol{\eta_B}\rangle_{\boldsymbol{B}} \;:=\; \mathrm{Tr}\big(\boldsymbol{B}^{-1}\boldsymbol{\zeta_B}\boldsymbol{B}^{-1}\boldsymbol{\eta_B}\big)$$

where $\boldsymbol{\eta},\boldsymbol{\zeta} \in T_{\boldsymbol{B}}\mathbb{S}_{++} = \{\boldsymbol{\eta} = \boldsymbol{\eta}^\top\}$ and with closed-form Riemannian exponential and logarithm:

$$\mathrm{Exp}_{\boldsymbol{B}}(\boldsymbol{\eta}) = \boldsymbol{B}^{1/2}\exp\big(\boldsymbol{B}^{-1/2}\boldsymbol{\eta}\boldsymbol{B}^{-1/2}\big)\boldsymbol{B}^{1/2}\,, \tag{25a}$$
$$\mathrm{Log}_{\boldsymbol{B}}(C) = \boldsymbol{B}^{1/2}\log\big(\boldsymbol{B}^{-1/2}C\boldsymbol{B}^{-1/2}\big)\boldsymbol{B}^{1/2}\,, \tag{25b}$$

where $\exp(\cdot)$ denotes the matrix exponential. For the Stiefel factors, we deviate from Mishra et al. (2013) and use the canonical Stiefel metric

$$\langle\boldsymbol{\zeta_U},\boldsymbol{\eta_U}\rangle_{\boldsymbol{U}} = \mathrm{Tr}\left(\boldsymbol{\zeta}_{\boldsymbol{U}}^\top(\boldsymbol{I} - \frac{1}{2}\boldsymbol{U}\boldsymbol{U}^\top)\boldsymbol{\eta_U}\right)$$

which has a better-behaved numerical routine for calculating its Log (using the algorithm from Mataigne et al. (2024)). Edelman et al. (1998) derived the exponential map: Given a point $\boldsymbol{U} \in \mathrm{St}(n, r)$ and a tangent vector $\boldsymbol{\eta_U} \in T_{\boldsymbol{U}}\mathrm{St}(n, r)$, first create the compact QR factorization $(\boldsymbol{I} - \boldsymbol{U}\boldsymbol{U}^\top)\boldsymbol{\eta} = \boldsymbol{Q}\boldsymbol{R}$, with $\boldsymbol{Q} \in \mathrm{St}(n, r),\;\; R \in \mathbb{R}^{r\times r}$. Then, $\boldsymbol{A} := \boldsymbol{U}^\top\boldsymbol{\eta}$ is skew-symmetric, i.e., $\boldsymbol{A}^\top = -\boldsymbol{A}$. With these definitions, the exponential mapping is

$$\mathrm{Exp}_{\boldsymbol{U}}(\boldsymbol{\eta}) \;=\; \begin{pmatrix}\boldsymbol{U} & \boldsymbol{Q}\end{pmatrix}\exp\left(\begin{pmatrix}\boldsymbol{A} & -\boldsymbol{R}^\top \\ \boldsymbol{R} & 0\end{pmatrix}\right)\begin{pmatrix}\boldsymbol{I}_p \\ 0\end{pmatrix} \tag{26}$$

and becomes particularly efficient when $r < n/2$ which is always the case for the LoRA adapters we will study—typically $(n = 4096) \times (r = 16)$ matrices.

The projection onto the horizontal space is given by

$$\eta^{\mathrm{hor}} = (\eta_{\boldsymbol{U}} - \boldsymbol{U}\boldsymbol{\Omega},\; \eta_{\boldsymbol{B}} - (\boldsymbol{B}\boldsymbol{\Omega} - \boldsymbol{\Omega}\boldsymbol{B}),\; \eta_{\boldsymbol{V}} - \boldsymbol{V}\boldsymbol{\Omega}),$$

with the skew-symmetric $\boldsymbol{\Omega}$ as numerical solution to the equation

$$\boldsymbol{B}^{-1}\boldsymbol{\Omega}\boldsymbol{B} + \boldsymbol{B}\boldsymbol{\Omega}\boldsymbol{B}^{-1} - \boldsymbol{\Omega} = \frac{1}{2}(\boldsymbol{V}^\top\eta_{\boldsymbol{V}} + \boldsymbol{U}^\top\eta_{\boldsymbol{U}}) - (\boldsymbol{B}^{-1}\eta_{\boldsymbol{B}} - \eta_{\boldsymbol{B}}\boldsymbol{B}^{-1})\,.$$

D. GeoMerge Implementation Details

This appendix gives the implementation-level details omitted from the main text. Lifted GeoMerge has two components: a direct endpoint lift from rank r to rank R, followed by a quotient Fréchet mean on the lifted points. The Fréchet solver below is used only in this second stage. We write a point in the solver manifold as $p = [U, B, V] \in \mathcal{M}_k$, with $k = R$ for the lifted experiments.

The only subtlety is that the logarithm $\mathrm{Log}_p(q)$ must compare representatives in a common gauge. Thus each Fréchet

Algorithm 1 Quotient Fréchet Mean with Orbit Alignment

Require: Quotient points $\{q_i = [U_i, B_i, V_i]\}_{i=1}^{T} \subset \mathcal{M}_k$, weights $\{w_i\}_{i=1}^{T}$, initializer p_0, step size α, maximum iterations N, first and later alignment iterations A_1, A_+, tolerance ε
Ensure: Fréchet mean estimate $\mu \in \mathcal{M}_k$

$p \leftarrow p_0$; initialize $O_i \leftarrow \mathrm{polar}(U_i^\top U_p + V_i^\top V_p)$ for all i ▷ one-time Procrustes
for $n = 1$ to N **do**
  $A_n \leftarrow A_1$ if $n = 1$, otherwise A_+
  for $i = 1$ to T **do**
    for $a = 1$ to A_n **do**
      $\tilde{q}_i \leftarrow q_i \cdot O_i$
      $\zeta_i \leftarrow \mathrm{Log}_p^{\mathrm{total}}(\tilde{q}_i)$
      $\Omega_i \leftarrow \mathrm{GaugeDrift}(p, \zeta_i)$ ▷ skew drift from horizontal equation
      $O_i \leftarrow O_i \exp(-\tau\Omega_i)$
    $\tilde{q}_i \leftarrow q_i \cdot O_i$
    $\eta_i \leftarrow \Pi_p^H \mathrm{Log}_p^{\mathrm{total}}(\tilde{q}_i)$ ▷ horizontal quotient log
  $\bar{\eta} \leftarrow \sum_{i=1}^{T} w_i \eta_i$
  if $\|\bar{\eta}\|_p < \varepsilon$ **then**
    return p
  $p \leftarrow \mathrm{Exp}_p(\alpha\bar{\eta})$
return p

Algorithm 2 Lifted GeoMerge

Require: Rank-r quotient points $\{\theta_t = [U_t, B_t, V_t]\}_{t=1}^{T} \subset \mathcal{M}_r$, target rank $R > r$, weights $\{w_t\}_{t=1}^{T}$
Ensure: Rank-R merged point $\mu_R \in \mathcal{M}_R$

$c \leftarrow \left(\prod_{t=1}^{T} \lambda_{\min}(B_t)\right)^{1/T}$
for $t = 1$ to T **do**
  $\mathcal{P}_t \leftarrow \{s : s \neq t\}$
  for $s \in \mathcal{P}_t$ **do**
    $A_{ts} \leftarrow U_s - U_t(U_t^\top U_s)$
    $C_{ts} \leftarrow V_s - V_t(V_t^\top V_s)$
  $A_t \leftarrow [A_{ts}]_{s\in\mathcal{P}_t}, \quad C_t \leftarrow [C_{ts}]_{s\in\mathcal{P}_t}$
  $M_t \leftarrow \mathrm{blkdiag}(B_s)_{s\in\mathcal{P}_t}$
  $A_t = E_t^U G_t, \qquad C_t = E_t^V H_t.$
  $K_t \leftarrow G_t M_t H_t^\top$
  Compute $K_t = \bar{U}_t \Sigma_t \bar{V}_t^\top$
  $U_t^\perp \leftarrow E_t^U \bar{U}_t, \quad V_t^\perp \leftarrow E_t^V \bar{V}_t$
  Keep the leading $R - r$ columns of $U_t^\perp$ and $V_t^\perp$
  $\widehat{U}_t \leftarrow [U_t, U_t^\perp], \quad \widehat{V}_t \leftarrow [V_t, V_t^\perp]$
  $\widehat{B}_t \leftarrow \begin{bmatrix} B_t & 0 \\ 0 & cI_{R-r} \end{bmatrix}$
  $L_t \leftarrow [\widehat{U}_t, \widehat{B}_t, \widehat{V}_t] \in \mathcal{M}_R$
$\mu_R \leftarrow \mathrm{FrechetMean}_{\mathcal{M}_R}(\{L_t\}_{t=1}^{T}; \{w_t\}_{t=1}^{T})$ ▷ Algorithm 1
return μ_R

iteration first aligns every sample q_i to the current iterate p, then averages the resulting horizontal logarithms. For $q = [U_q, B_q, V_q]$, the $O(k)$ gauge action is

$$q \cdot O = [U_q O,\ O^\top B_q O,\ V_q O], \qquad O \in O(k).$$

The alignment step approximately solves

$$O^\star(p; q) \in \arg\min_{O\in O(k)} d_{\bar{\mathcal{M}}_k}\big(p, q \cdot O\big)^2,$$

where $\bar{\mathcal{M}}_k = \mathrm{St}(d_{\mathrm{out}}, k) \times \mathrm{SPD}(k) \times \mathrm{St}(d_{\mathrm{in}}, k)$ is the total space. In practice, each task gauge is initialized once by Procrustes at the first outer iterate and then retained across outer iterations. The Procrustes initialization aligns the Stiefel factors: $O_0 = \mathrm{polar}\big(U_q^\top U + V_q^\top V\big)$. Given a current gauge O, we form the raw total-space logarithm from p to $q \cdot O$ and solve the horizontal projection equation for the skew-symmetric drift Ω. We then update

$$O \leftarrow O \exp(-\tau\Omega),$$

for a small fixed number of inner iterations (we use 5 for the first outer iteration, 2 thereafter). The final logarithm returned to the Fréchet solver is the horizontal projection of the aligned raw logarithm.

Here Π_p^H denotes horizontal projection. For a raw tangent $\zeta = (\zeta_U, \zeta_B, \zeta_V)$ at $p = [U, B, V]$, this projection has the form

$$\Pi_p^H \zeta = \big(\zeta_U - U\Omega,\ \zeta_B - (B\Omega - \Omega B),\ \zeta_V - V\Omega\big),$$

where $\Omega^\top = -\Omega$ is obtained from the horizontal gauge equation

$$B^{-1}\Omega B + B\Omega B^{-1} - \Omega = \frac{1}{2}\left(V^\top \zeta_V + U^\top \zeta_U\right) - \left(B^{-1}\zeta_B - \zeta_B B^{-1}\right).$$

Thus $\mathrm{GaugeDrift}(p, \zeta)$ in Algorithm 1 is the skew matrix Ω solving this equation. In the final line, Exp_p is implemented as the product update on the total-space factors: a Stiefel update for U and V, and the affine-invariant SPD exponential for B.

Typically we use $\alpha = 1.0$ in our experiments since this seems to be stable.

We next describe the algorithm for the rank-lifted procedure. After constructing lifted points $L_t \in \mathcal{M}_R$, it computes the final rank-R quotient Fréchet mean.

E. Additional NLI Experiments

Following prior work (Stoica et al., 2025), for language tasks we consider six LoRA-adapted specialists derived from the

Table 2. Merged-model performance on NLI tasks for Llama-3 8B in normalized accuracies, the ratio of the accuracy of the merged model on each dataset to that of the original LoRA adapters.

Method	Framework	Normalized Accuracy (%)						Avg.
		SNLI	MNLI	SICK	QNLI	RTE	SciTail	
TA	*Standard*	93.57	95.28	87.96	68.71	100.0	96.73	90.38
	GeoMerge	91.416	92.159	90.274	83.749	100.00	93.665	91.88

same Llama 3 8B base model (Grattafiori et al., 2024), each fine-tuned on a different NLI dataset: SNLI (Bowman et al., 2015), MNLI (Williams et al., 2018), SICK (Marelli et al., 2014), QNLI (Wang et al., 2018), RTE (Wang et al., 2018), and SciTail (Khot et al., 2018).

Table 2 summarizes NLI merging results on Llama 3 8B: per-task normalized accuracy and average normalized accuracy. GeoMerge outperforms the baseline in this setting. We did not run our full stack in this experimental setting due to computational limitations of our available Titan V GPU. We expect the full stack to perform considerably better, since Panariello et al. (2025) shows that restricting rank to the original rank r has a severe effect on merged model performance.

F. Infrastructure Details

Compute Time and Resources. Our merging algorithm runs on GPU and takes less than 1 hour on a machine with a Titan V GPU, with Intel(R) Xeon(R) W-2133 CPU 12 cores @ 3.60GHz.

In terms of compute time for the NLI datasets, on our machine, TA took 9s, KnOTS around 70 mins, GeoMerge at rank 16 with exact Exp/Log around 2hrs, and GeoMerge with Cayley retractions/logs at rank 16 around 60 mins. Lifted GeoMerge with Cayley retractions/log on the vision datasets took around 20 mins per run

Dataset Licenses. Among the datasets we use, we were able to determine the following licenses and/or usage permissions:

- Cars (Krause et al., 2013) uses Creative Commons License.
- GTSRB (Stallkamp et al., 2011) uses Creative Commons License.
- EuroSAT (Helber et al., 2019) uses MIT license.
- MNIST (LeCun, 1998) uses Gnu General Public License and elsewhere is listed under Creative Commons Attribution-Share Alike 3.0 license.
- SUN397 (Xiao et al., 2016) is listed as “for research purposes only”.
- SVHN (Netzer et al., 2011) is listed as being “for non-commercial use only”.
- SNLI (Bowman et al., 2015) uses CC BY-SA 4.0
- MNLI (Williams et al., 2018) appears to incorporate an aggregate of multiple licenses.
- SICK (Marelli et al., 2014) uses a Creative Commons Attribution-NonCommercial-ShareAlike license.
- QNLI (Wang et al., 2018) is derived from SQuAD, which in turn uses a CC BY-SA 4.0 license.
- RTE (Wang et al., 2018) appears to incorporate an aggregate of multiple licenses.
- SciTail (Khot et al., 2018) uses an Apache 2.0 license.

G. Additional Related Work

Weight averaging and basin structure. Model soups (Wortsman et al., 2022) show that averaging fine-tuned checkpoints can improve accuracy without inference-time overhead. A common explanation invokes flatness and (approximate) linear mode connectivity (Garipov et al., 2018; Draxler et al., 2018), which makes Euclidean averaging benign when solutions live in one convex-like region. GeoMerge does not contradict this; rather, it asks what happens when Euclidean geometry is *not* the right notion of closeness—a question that becomes especially pressing in the presence of symmetries.

Symmetries, quotient geometry, and deep learning. Weight-space symmetries are known to invalidate naive Euclidean notions of sharpness and flatness (Dinh et al., 2017). da Silva et al. (2025) study the high-dimensional $\mathrm{GL}(h)$ symmetries in transformers' attention heads, and develop symmetry-corrected sharpness measures on quotient manifolds (Boumal, 2023). We extend their recipe by studying Riemannian averaging operators and target *model merging* rather than sharpness.

KnOTS and LoRA. KnOTS (Stoica et al., 2025) improves LoRA merging via joint SVD-based transformations to better align low-rank updates before applying existing merge rules. GeoMerge complements this; it lets us see SVD-based alignment as a particular gauge fixing for $\mathrm{GL}(r)$ symmetries of low-rank factorizations, selecting comparable representatives of the same underlying updates before averaging.